

Channel 1: Expert Maps

Curriculum Trees & Prereqs
(PDKT-C, CMKT, HKT)

Channel 2: Co-occurrence

Interaction Bipartite Graphs
(GKT, GIKT, Bi-CLKT)

Channel 3: Dynamic Streaming

Time-Evolving Interaction Windows
(DyGKT, STG-SKT)

Channel 4: LLM Syntax

AST & Code Semantic Trees
(SINKT, Coda, KGNN-KT)

Graph Encoder Processing

GCN / Hetero-GNN / Hyperbolic
Neighborhood Message Passing

Concept Adjacency Matrix A

& Topological Embeddings Z
For Downstream KT Prediction

